



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

B. Sc. CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	PERSPECTIVES OF INDUSTRIAL CHEMISTRY				
Type of Course	VOCATIONAL MINOR				
Semester	II				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Elementary ideas of fuel, food, fertilizers, cleansing agents, dyes and cosmetics. 2. Basic understanding of volumetric titrations				
Course Summary	This course details the processes involved in fuel production, manufacturing and chemistry involved in cleansing agents, dyes, cosmetics and various fertilizers and pesticide industries. The course also furnishes practical knowledge about the preparation of the above described products.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Learn the composition and refining processes of crude petroleum and petroleum products	U	F	Instructor-created exams / Group Tutorial Work
CO2	Gain insight into non-petroleum fuels and clean fuels, various lubricants and industrial usage of coal	R	F	Instructor-created exams / Home assignments
CO3	Acquire knowledge in food chemistry which includes an overview on various food additives such as food colours, flavours and artificial sweeteners	U	C	Group Tutorial Work/Seminar
CO4	Analyse the composition and functioning of various cleansing agents, dyes and cosmetics	An	P	Instructor-created exams / Viva
CO5	Analyse the different types of pesticides as insecticides, herbicides, rodenticides and fungicides	An	M	Instructor-created exams / Assignments

CO6	Gain practical skills in some industrially important chemical analyses including test for food adulteration, acid value, ester value and saponification value of oils, TFM of soaps and preparation of some cosmetic products	Ap	P	Practical Assignment / Observation of Practical Skills
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Fuel Chemistry		9	20
	1	Review of energy resources (renewable and non-renewable -Clean Energy). Classification of fuels and their calorific value.	2	
	2	Uses of coal (fuel and non fuel) in various industries and its composition	1	
	3	Composition of crude petroleum, Refining and different types of petroleum products and their applications.	1	
	4	Fractional Distillation (Principle and process), Cracking (Thermal and catalytic cracking)	1	
	5	Elementary ideas on Non-petroleum fuels (CNG, LNG, bio-gas, fuels derived from biomass), clean fuels (wind, solar, tidal)	2	
	6	Lubricants: Solid and semisolid lubricants, synthetic lubricants	2	
II	Food Chemistry		12	26
	7	Food additives: Functional food additives, Food colours-permitted and non permitted – adulteration and Toxicology.	3	
	8	Flavours – natural and synthetic – Soft drinks – formulation. Health drinks.	3	
	9	Artificial sweeteners – Artificial ripening of fruits and its side effects.	1	
	10	Modern Food Habits: Definition and health effects of fast foods, instant foods, dehydrated foods and junk foods. Harmful effects of modern food habits.	3	
	11	Spices: Introduction, (turmeric, chilli, coriander, pepper, cardamom, cloves), general extraction procedure, applications	2	
III	Cleansing agents, Dyes and Cosmetics		18	39
	12	Cleansing Agents: Soaps – Hard and soft soaps – Alkali content – TFM – Detergents (classification) – Cleaning action – Advantages and disadvantages of soaps and detergents	3	
	13	Shaving creams-Ingredients and functions, Shampoos – Different kinds of shampoos (Anti-dandruff, anti-lice, herbal and baby shampoos).	2	

	14	Tooth paste: composition and preparation.	1	
	15	Dyes: Definition – Requirements of a dye – Theories of colour and chemical constitution – Classification based on structure and mode of application to the fabric . Industrial method of dyeing	3	
	16	Cosmetics: Hair dye: Types, chemicals used and its harmful effects.	2	
	17	Face and skin creams: Types, ingredients and functions. Antiperspirants, sun screen, Face and skin powders, nail polishes, lipsticks, eyebrow pencils and eye liners (ingredients and functions) – Harmful effects of cosmetics	3	
	18	Perfumes: Science of smell, history of fragrance, fragrance sources, natural products and aroma chemicals used in fragrances,	2	
	19	Fragrance applications in personal care and household products, safety and regulations of fragrance, emotional and psychological effects of odours.	2	
IV	Fertilizers and Pesticides		6	13
	20	Fertilizers: Essential nutrients for plants – NPK value – Natural and synthetic fertilizers	2	
	21	Nitrogenous, phosphatic and potash fertilizers (examples) – Impact of excessive use of fertilizers on environment – Biofertilizers.	2	
	22	Pesticides: Classification – Insecticides, herbicides, rodenticides and fungicides (definition and examples only) – Non-degradable pesticides	2	
V	Practical		30	
	1	Test for adulteration in selected food products (5 experiments)	15	
	2	Determination of Acid value, ester value and saponification value of oils, TFM of soaps	7	
	3	Preparation of pain balms, lip balms, soaps, liquid detergent, shampoo, hand sanitizer and disinfectants	8	

Books and References:

1. B.K. Sharma, Industrial Chemistry, Goel Publications (1983).
2. R.K. Das, Industrial Chemistry, Kalyani Publications, New Delhi (1982).
3. G. R. Chatwal, Synthetic Drugs, Himalaya Publishing House, Bombay, 1995.
4. J.Ghosh, A Textbook of Pharmaceutical Chemistry, S. Chand & Co Ltd., 1997
5. B. Sreelakshmi, Food Science, New Age International Pvt. Ltd, New Delhi, 2015
6. D. Swern, Bailey's Industrial Oil and Fat Products, Vol. I and II, 4th Edn., John Wiley, 1982.
7. A. I. Vogel 'A Text Book of Quantitative Inorganic Analysis Including Elementary Instrumental Analysis': (Third Ed.) (ELBS)

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	-	-	-	-	-	2	3	-	2	-	3	-	3
CO 2	-	-	-	-	-	2	3	-	2	-	3	2	3
CO 3	-	-	-	-	-	2	3	-	2	-	3	-	3
CO 4	-	-	-	-	-	2	3	-	2	1	3	-	3
CO 5	-	-	-	-	-	2	3	-	2	1	3	-	3
CO 6	-	-	2	-	-	2	3	-	2	-	3	-	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment / Discussion / Seminar
- Internal Examination
- Practical Examination
- Final Examination (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3		✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓		✓	